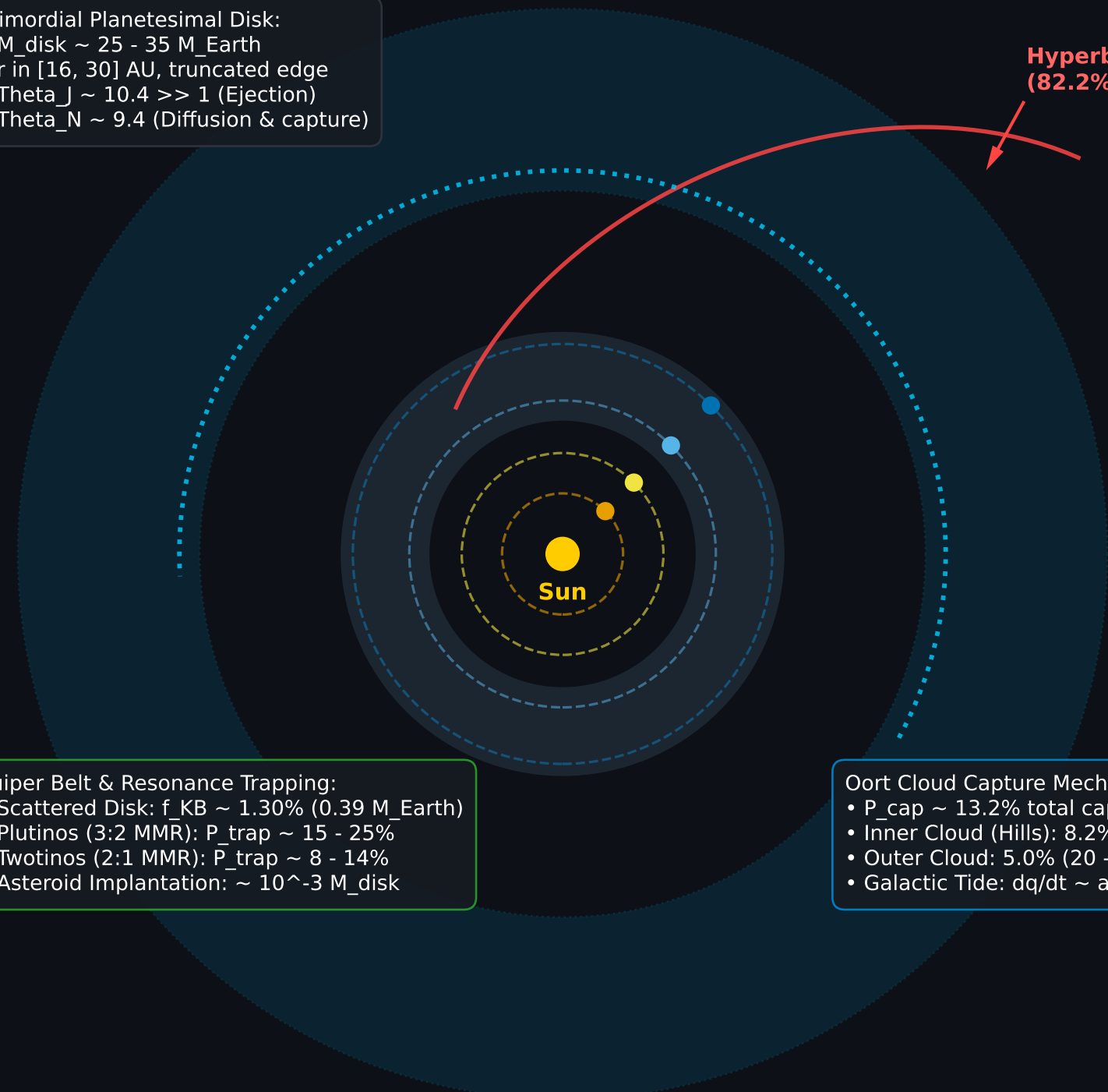


Dynamical Architecture of Planetesimal Scattering, Kuiper Belt Implantation & Oort Cloud Trapping

- Primordial Planetesimal Disk:
- $M_{\text{disk}} \sim 25 - 35 M_{\text{Earth}}$
 - r in [16, 30] AU, truncated edge
 - $\Theta_J \sim 10.4 \gg 1$ (Ejection)
 - $\Theta_N \sim 9.4$ (Diffusion & capture)



- Kuiper Belt & Resonance Trapping:
- Scattered Disk: $f_{\text{KB}} \sim 1.30\%$ ($0.39 M_{\text{Earth}}$)
 - Plutinos (3:2 MMR): $P_{\text{trap}} \sim 15 - 25\%$
 - Twotinos (2:1 MMR): $P_{\text{trap}} \sim 8 - 14\%$
 - Asteroid Implantation: $\sim 10^{-3} M_{\text{disk}}$

- Oort Cloud Capture Mechanics:
- $P_{\text{cap}} \sim 13.2\%$ total capture
 - Inner Cloud (Hills): 8.2% (2 - 20 kAU)
 - Outer Cloud: 5.0% (20 - 50 kAU)
 - Galactic Tide: $dq/dt \sim a^{3.5} \Rightarrow q > 32 \text{ AU}$